

“Thank you, Deniz. The core problem: the kits wait on the reach truck, and that waiting is most of the lead time, not the walking. My job was to turn that into a model we can experiment on, without moving a real pallet. So we rebuilt the whole kitting process as a discrete-event simulation in Python, using SimPy.”

“This diagram is the whole pipeline. Orders arrive in batches, exactly like the real dispatch log. For every material, the model looks up its location, set by the slotting policy we test. Then: Kardex, a carousel brings it; low and hand-reachable, an operator picks it; up high, wait for one of the seven reach trucks. The operator builds the kit, walks it to the line, and we record the KPIs. Six policies, twenty replications each.”

“This is what makes it credible: the model is distribution-driven, not a replay. From four months of the dispatch log, over 160,000 rows, I fit the arrivals, the kit sizes, the kit contents per line, and the pick times, then sample from them. Because demand is generated, not fixed, I can re-slot the materials and re-run the exact same warehouse under any policy, and the comparison stays fair. The pick-time variability comes straight from the F400 video study, and all six policies share common random numbers.”

“And this is not a picture of the model, this is the model, running live, from overhead. Follow one operator: the picks stay low, on the manual shelves next to the line, and the reach truck is almost never called. That absence, the truck that does not move, is the entire result, and every number after me comes from this engine. I will now hand over to Nehir.”

Q1. Trace replay or a real model? Distribution-driven, not a replay. I fit distributions from the 4-month log and sample, so I can re-slot and re-run any policy. Python + SimPy.

Q2. Why discrete-event simulation / SimPy? The bottleneck is queueing for shared resources and the slot-hold cascade, which static models miss. SimPy makes every resource explicit.

Q3. Where do the pick times come from? The F400 video study, 2,319 micro-events. Lognormal pick times, with the spread taken from the measured CV.

Q4. Kardex timing, no Kardex video? Its mean and CV mirror the operator pick, disclosed. It cannot bias the result: Kardex routing is identical across all six policies.

Q5. F400 extrapolated to nine families? F400 is about 29% of rows; a $\pm 20\%$ sensitivity sweep keeps Line-aware first under every swing. Per-line studies are future work.

Q6. Why batch arrivals, does the rate match? The log arrives in bursts: 3,885 batches, mean 4.68 kits per batch. The runs land about 400 a day, within 1.1% of the real 396.

Q7. Why 20 replications, same seeds? Seed $42+i$; common random numbers give every policy the same demand. The 95% CIs are narrow and non-overlapping, so 20 is enough.

Q8. Why six policies, not only yours? To benchmark fairly: SAP baseline, a heuristic control, two ABC variants, travel-distance, and Line-aware. Without references, “better” means nothing.

Q9. Chi-square significant, model wrong? No. $\chi^2(7) = 535$, but Cramer's V is only 0.22. Significant because the model uses its own routing, not a replay; read the effect size.

Q10. No holdout, same data for fit and validation? Correct and disclosed: SAP logged no waits or RT calls. Grounding: F400 video, daily volume within about 1%, zero rack crossings.

Q11. Aisle congestion not modelled? Absolute times are slightly optimistic, but all six policies share it symmetrically, so the ranking holds. It would only widen Line-aware's lead.

Q12. How real is the SAP baseline? An honest SAP-plus-heuristic hybrid: 750 at exact SAP bins, 2,872 Kardex, the rest heuristic. Not a pure replay, but our ground truth.

Q13. Throughput flat, RT only 1.3%? Demand-limited: same arrivals, so about 398 a day for all. The 1.3% RT is picking only, fair across all six; the live total would be higher.

Q14. What about the milk-run trains? Seven are defined as a transport mode, but the operator carries the kit in these runs. Enabling it offloads walking, not the reach-truck queue.

Q15. Is the warehouse layout real? Reconstructed from the Auto-CAD drawing and 11 rack PDFs, blind-CAD checked. 3,101 modelled positions reconcile with the 3,203 drawn.

Q16. Can someone reproduce your results? Yes: fixed-seed and deterministic; 6 policies $\times$ 20 replications = 120 runs, exported for Minitab.